

Hierarchical Parzen ICA for Neural Imaging

Background Reconstruction, Noisy Source Separation, and Stable Real-Time Inference

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Abstract

NeuRev experiments have established two complementary facts. First, adjacent-frame InfoMax and Cauchy–Schwarz Parzen separation recover a direction nearly identical to temporal subtraction, showing that pairwise ICA naturally separates a persistent common mode from change. Second, amplitude-preserving representations are currently more useful than derivative-only features: an offline stable AR(1) smoother and low-rank amplitude PCA improve known-label recovery, whereas raw and latent differences, model-aware dynamic drive, and filter innovation underperform. These results motivate a hierarchical rather than monolithic source model. Stage 1 uses batch or stochastic Parzen ICA on aligned frame pairs to identify and reconstruct a background-like common component while preserving an amplitude residual. Stage 2 treats overlapping residual patches as noisy linear mixtures, estimates observation-noise covariance from quiet residuals, performs noise-corrected subspace selection, and fits a local noisy Parzen ICA model. The source density is represented by a Gaussian Parzen dictionary convolved with the projected additive-noise distribution; this yields both a noise-aware score function for stochastic demixing updates and an analytic posterior mean for clean-source estimation. Structured components are subsequently classified as neural signal, artifact, or unresolved using spatial localization, annularity, temporal coherence, recurrence, motion correlation, and stability. We specify identifiability limitations, numerical and scientific stability requirements, a real-time two-timescale architecture, a falsification-oriented synthetic and semi-synthetic program, and mandatory visual and quantitative outputs centered on decomposition closure, background–signal–artifact–noise leakage, event preservation, residual validity, downstream detection, and latency.

1 Canonical problem statement

The desired scientific result is not simply a smooth movie. It is a decomposition whose channels have distinct meanings:

$$Y_t(p) = B_t(p) + S_t(p) + A_t(p) + N_t(p), \quad (1)$$

where B is background-like structure, S is accepted neural dynamics, A is structured artifact, and N is stochastic measurement noise. A user-facing three-way view may combine $A + N$ as residual uncertainty, but the internal four-way representation is necessary: motion edges, saturation, and coherent acquisition errors are not ordinary noise and should not be promoted to neural signal.

The hierarchy proposed here is

$$Y \xrightarrow{\text{Stage 1}} \hat{B} + R, \quad R \xrightarrow{\text{Stage 2}} \hat{S} + \hat{A} + \hat{N}. \quad (2)$$

The end-to-end closure is

$$E_{\text{closure}} = Y - \hat{B} - \hat{S} - \hat{A} - \hat{N}. \quad (3)$$

A decomposition is not credible unless E_{closure} is small and the energy assigned to each channel is scientifically appropriate.

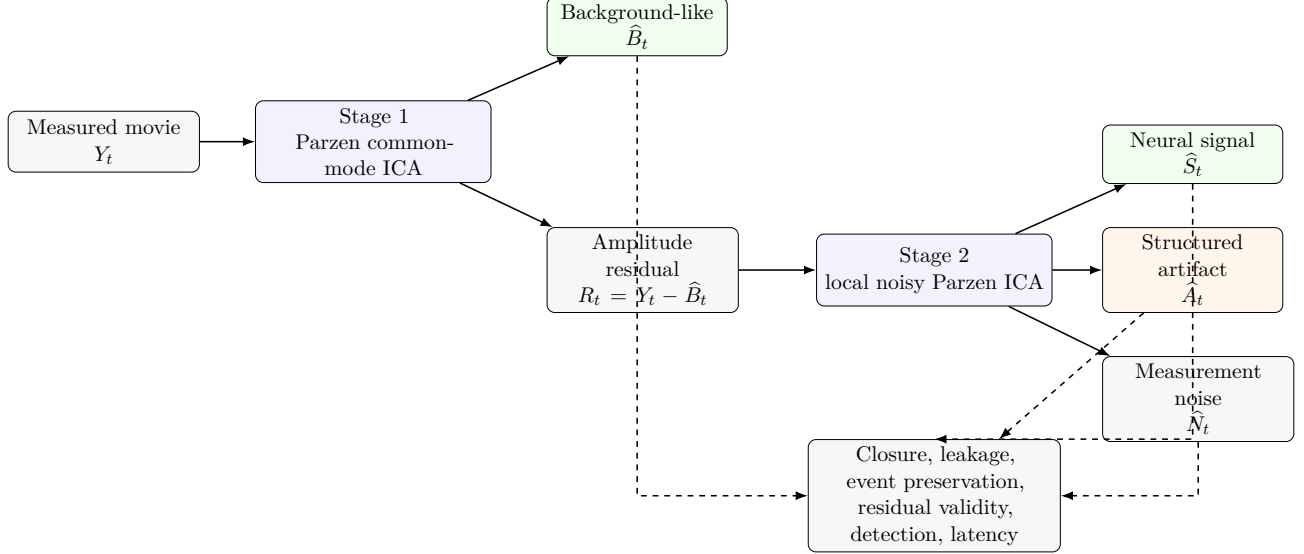


Figure 1: The proposed scientific order. Stage 1 reconstructs rather than merely forwards a derivative component. Stage 2 explicitly models additive noise and retains coherent non-neural residuals as artifacts.

2 What the present NeuRev evidence implies

The current repository provides four motivating observations.

Pairwise ICA became a derivative. For adjacent samples

$$\mathbf{x}_t(p) = \begin{bmatrix} I_{t-1}(p) & I_t(p) \end{bmatrix}^T, \quad (4)$$

persistent structure lies near $[1, 1]^T$ and its orthogonal change direction is $[-1, 1]$. Existing InfoMax and CS-Parzen fits recovered effective directions almost collinear with subtraction. This is useful evidence that the pair contains a common/background-like mode, but it does not identify every physical source.

Change alone did not replace amplitude. Fixed/adaptive differences and pairwise ICA were useful as onset or change evidence but did not improve the frozen Raw Direct detector under the tested additive and gating rules. Strong gates removed slowly evolving calcium activity.

Denoised amplitude helped. A stable shared AR(1) offline smoother improved known-label recovery over Raw Direct, while raw and latent differences, positive dynamic drive, and standardized filter innovation underperformed. The useful representation was latent amplitude, not post-denoising differencing.

Low rank helped only modestly and high-rank ICA destabilized. Amplitude PCA at rank eight produced a small provisional fixed-budget gain. Rank-16 spatial ICA was stable but did not win at fixed budget; rank-64 ICA hit the iteration cap and showed poor seed stability. This argues for bounded local rank and explicit stability gates.

The new hierarchy therefore preserves the common-versus-change idea, but the Stage 1 output is an amplitude residual, and the Stage 2 model is local, noise-aware, and rank constrained.

3 Identifiability boundary

The decomposition in eq. (1) is not uniquely identifiable without structural assumptions. Two limitations matter immediately.

3.1 Two observations identify at most two aggregate directions

For one pixel and two adjacent frames, the data contain background, previous and current neural state, previous and current noise, and possible motion. A 2×2 ICA can recover at most two aggregate components. Stage 1 should therefore be interpreted as

$$\text{common/background-like mode} \quad | \quad \text{dynamic residual mode}, \quad (5)$$

not as a complete physical decomposition.

3.2 Noise is not an ordinary ICA source

For a noisy mixture

$$\mathbf{r}_t = A\mathbf{s}_t + \mathbf{n}_t, \quad (6)$$

Gaussian noise does not generally define a unique ICA direction. After whitening, isotropic Gaussian noise is rotation invariant. The correct Stage 2 task is to identify reproducible structured sources under an additive-noise model and leave the remaining unexplained variation as residual uncertainty. Noisy-ICA theory also distinguishes estimation of the mixing matrix from minimum-noise source recovery; naive inversion is not automatically optimal [6].

4 Stage 1: Parzen common-mode separation

4.1 Pair construction and gain correction

For lag k , define

$$\mathbf{x}_t(p) = \begin{bmatrix} I_{t-k}(p) \\ I_t(p) \end{bmatrix}. \quad (7)$$

The initial reference uses $k = 1$. A robust gain model may estimate

$$I_t(p) \approx \alpha_t I_{t-k}(p) + d_t(p), \quad (8)$$

so that the gain-adjusted common and differential directions are approximately

$$\mathbf{c}_t = \begin{bmatrix} 1 \\ \alpha_t \end{bmatrix}, \quad \mathbf{d}_t = \begin{bmatrix} -\alpha_t \\ 1 \end{bmatrix}. \quad (9)$$

The gain coefficient α_t is an observation parameter and must not be confused with a calcium decay coefficient.

4.2 Stable whitening

Center and whiten bounded spatial samples:

$$\mathbf{z}_t(p) = Q_t[\mathbf{x}_t(p) - \boldsymbol{\mu}_t]. \quad (10)$$

The covariance eigendecomposition must use a floor and report its condition number. If the second eigenvalue is below the declared floor, the pair is unidentifiable and no arbitrary ICA component should be emitted.

4.3 Batch CS-Parzen reference

The existing NeuRev reference estimates independence with a Gaussian-Parzen Cauchy–Schwarz divergence. For densities p and q ,

$$D_{CS}(p, q) = -\log \frac{(\int p(u)q(u) du)^2}{\int p(u)^2 du \int q(u)^2 du}. \quad (11)$$

For two whitened outputs, the demixer is an orthogonal rotation, so a bounded angle search supplies an auditable reference. Kernel work must use deterministic subsamples and blockwise evaluation.

4.4 Stochastic Parzen score-function ICA

Stochastic information-gradient methods were introduced to reduce the cost of online entropy manipulation from all-pairs estimation toward bounded stochastic updates [2]. For component j , maintain a dictionary $\{c_{jm}\}_{m=1}^M$ and Gaussian bandwidth h_j :

$$\hat{p}_j(y) = \frac{1}{M} \sum_{m=1}^M \mathcal{N}(y; c_{jm}, h_j^2). \quad (12)$$

The posterior responsibility of center m is

$$\alpha_{jm}(y) = \frac{\mathcal{N}(y; c_{jm}, h_j^2)}{\sum_{\ell} \mathcal{N}(y; c_{j\ell}, h_j^2)}, \quad (13)$$

and the negative score is

$$\psi_j(y) = -\frac{\partial}{\partial y} \log \hat{p}_j(y) = \frac{y - \sum_m \alpha_{jm}(y) c_{jm}}{h_j^2}. \quad (14)$$

A natural-gradient-style update is

$$W^+ = \text{decorrelate} \left(W + \eta [I - \boldsymbol{\psi}(\mathbf{y})\mathbf{y}^\top] W \right), \quad (15)$$

where the implementation must freeze a sign convention and verify the update against finite differences and the batch objective on small fixtures. Hild, Erdogmus, and Principe analyzed modified stochastic estimators that accommodate both super- and sub-Gaussian sources and exploit temporal diversity [4].

The dictionary is bounded. New centers should be posterior or filtered source samples, not an unbounded history of every observation. Minimum separation, replacement policy, age, usage, and bandwidth bounds must be recorded.

4.5 Identifying the background-like component

A near-zero *mean* derivative is insufficient because zero-mean noise also has that property. Use normalized derivative energies:

$$E_1(j) = \frac{\text{median}_{t,p}[(y_{j,t}(p) - y_{j,t-1}(p))^2]}{\text{Var}_{t,p}(y_{j,t}(p)) + \epsilon}, \quad (16)$$

$$E_2(j) = \frac{\text{median}_{t,p}[(y_{j,t}(p) - 2y_{j,t-1}(p) + y_{j,t-2}(p))^2]}{\text{Var}_{t,p}(y_{j,t}(p)) + \epsilon}. \quad (17)$$

For independent white noise with variance σ^2 ,

$$\text{Var}(\Delta n_t) = 2\sigma^2, \quad \text{Var}(\Delta^2 n_t) = 6\sigma^2, \quad (18)$$

so derivative energy separates static structure from noise more effectively than mean derivative.

The component score should also use:

- cosine similarity to the common and differential directions;
- spatial total variation or high-frequency mass;
- spatial support/broadness;
- correlation with global frame intensity;
- continuity across updates;
- event-versus-quiet modulation for reporting, not held-out selection.

A small score margin must return **unresolved**. This protects slow calcium activity, which may have low derivative energy over adjacent frames.

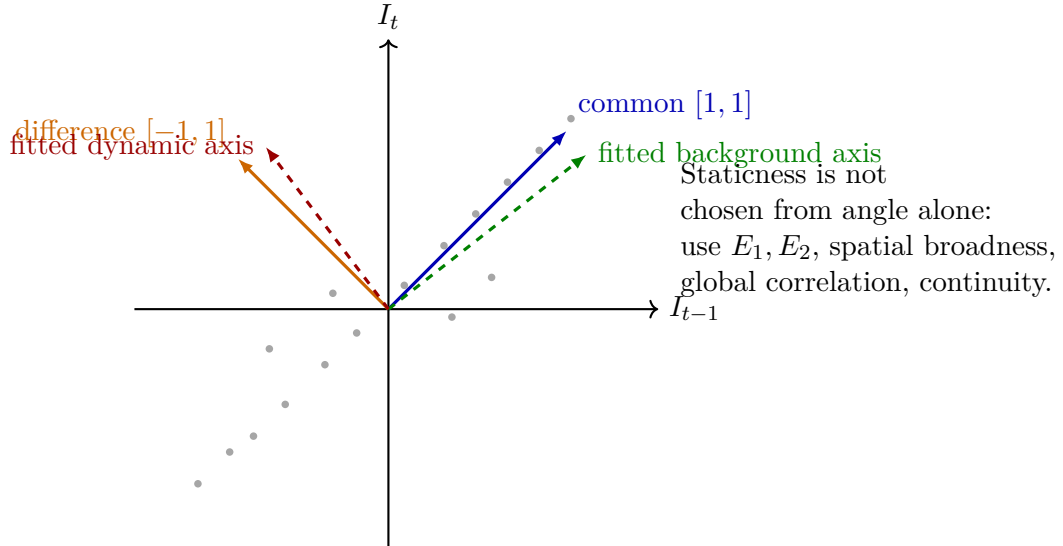


Figure 2: Stage 1 geometry. Persistent anatomy produces an elongated common mode; change and noise occupy the orthogonal direction. The actual component label must combine geometry with temporal and spatial diagnostics.

4.6 Background reconstruction and amplitude residual

Let

$$\mathbf{y} = WQ(\mathbf{x} - \boldsymbol{\mu}). \quad (19)$$

If W is orthogonal after whitening,

$$\mathbf{x} = \boldsymbol{\mu} + Q^\dagger W^\top \mathbf{y}. \quad (20)$$

Set the dynamic component to zero to form $\hat{\mathbf{x}}^{(B)}$. The current-frame background is

$$\hat{B}_t(p) = [\hat{\mathbf{x}}_t^{(B)}(p)]_2, \quad (21)$$

and the Stage 1 residual is

$$R_t(p) = I_t(p) - \hat{B}_t(p). \quad (22)$$

Equation (22) is central: Stage 2 receives an amplitude-preserving residual, not only the derivative-like ICA output.

5 Stage 2: local noisy Parzen ICA

5.1 Patchwise noisy mixture

For overlapping patch k with P pixels,

$$\mathbf{r}_t^{(k)} = A_k \mathbf{s}_t^{(k)} + \mathbf{a}_t^{(k)} + \mathbf{n}_t^{(k)}. \quad (23)$$

The patch provides multiple sensors; this is necessary because one scalar pixel trace cannot identify multiple latent sources and noise. Local patches also exploit neighboring-pixel coherence and bound the component rank.

5.2 Noise estimation and signal-subspace selection

Estimate quiet residual covariance $\hat{\Sigma}_n$ using a diagonal, shrinkage, or gated low-rank-plus-diagonal model. The structured covariance is

$$\hat{\Sigma}_s = \Pi_{\geq 0}(\hat{\Sigma}_r - \hat{\Sigma}_n), \quad (24)$$

where negative eigenvalues are clipped. Select only positive modes that exceed noise-floor and temporal-block stability thresholds. This is a required control: ordinary PCA whitening and noise-corrected whitening must be compared.

Let Q project and whiten the retained subspace:

$$\tilde{\mathbf{r}}_t = Q\mathbf{r}_t, \quad \mathbf{y}_t = W\tilde{\mathbf{r}}_t. \quad (25)$$

The projected noise covariance is

$$\Sigma_n^\sim = Q\hat{\Sigma}_n Q^\top. \quad (26)$$

For output row $\mathbf{w}_j^\top$,

$$\nu_j^2 = \mathbf{w}_j^\top \Sigma_n^\sim \mathbf{w}_j. \quad (27)$$

5.3 Noise-convolved Parzen density

Assume

$$y_j = s_j + \eta_j, \quad \eta_j \sim \mathcal{N}(0, \nu_j^2). \quad (28)$$

Represent the latent source density by

$$\hat{p}_{s_j}(s) = \frac{1}{M} \sum_{m=1}^M \mathcal{N}(s; c_{jm}, h_j^2). \quad (29)$$

Gaussian convolution gives the observed noisy density

$$\hat{p}_{y_j}(y) = \frac{1}{M} \sum_{m=1}^M \mathcal{N}(y; c_{jm}, h_j^2 + \nu_j^2). \quad (30)$$

The source bandwidth h_j and observation-noise variance ν_j^2 have different meanings and must not be collapsed into one unconstrained kernel parameter.

Define

$$\alpha_{jm}(y) = \frac{\mathcal{N}(y; c_{jm}, h_j^2 + \nu_j^2)}{\sum_{\ell} \mathcal{N}(y; c_{j\ell}, h_j^2 + \nu_j^2)}, \quad (31)$$

then

$$\psi_j(y) = -\frac{\partial}{\partial y} \log \hat{p}_{y_j}(y) = \frac{y - \sum_m \alpha_{jm}(y) c_{jm}}{h_j^2 + \nu_j^2}. \quad (32)$$

Substituting eq. (32) into the bounded natural update eq. (15) yields the proposed stochastic noisy Parzen ICA lane. An earlier adaptive-filter study used an augmented ICA model, Parzen density estimation, and stochastic information gradient to separate a signal from noisy observations by maximizing independence rather than minimizing mean-square error [5]. The present proposal differs in its spatial patch model, explicit noise-convolved source density, posterior source reconstruction, and four-way residual qualification.

5.4 Posterior clean-source estimate

For prior component m ,

$$s_j \sim \mathcal{N}(c_{jm}, h_j^2), \quad y_j = s_j + \eta_j. \quad (33)$$

Gaussian conditioning gives

$$\mu_{jm}(y) = \mathbb{E}[s_j \mid y_j = y, m] = \frac{\nu_j^2 c_{jm} + h_j^2 y}{h_j^2 + \nu_j^2}. \quad (34)$$

The mixture posterior mean is

$$\hat{s}_j(y) = \sum_m \alpha_{jm}(y) \mu_{jm}(y). \quad (35)$$

This is a principled denoised source amplitude. The Parzen dictionary should be updated with posterior source estimates, not raw noisy outputs, to avoid simply learning the noise-broadened distribution.

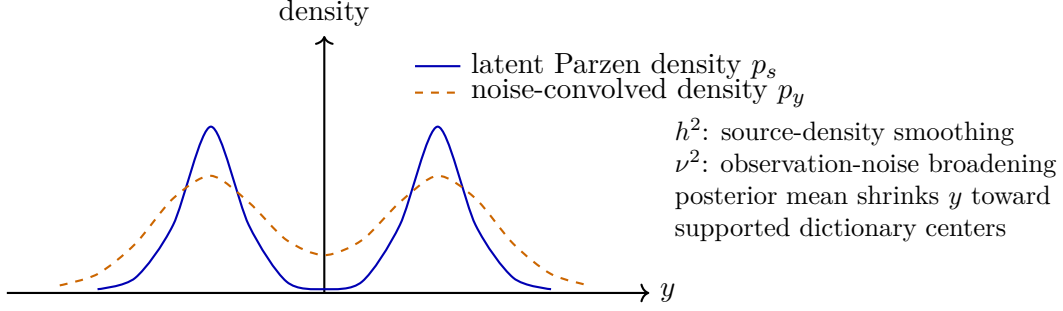


Figure 3: Schematic, not experimental data. Gaussian noise broadens the observed Parzen density analytically. The same model supplies the score function used for demixing and the posterior mean used for source denoising.

5.5 Patch reconstruction

Map the posterior sources back to patch coordinates:

$$\hat{\mathbf{r}}_t^{(S)} = Q^\dagger W^{-1} \hat{\mathbf{s}}_t. \quad (36)$$

For an orthogonal W , $W^{-1} = W^\top$; otherwise use a checked pseudoinverse and report its condition number. Overlapping patches are combined with a declared Hann or Tukey window and normalization.

The preliminary residual is

$$\mathbf{e}_t = \mathbf{r}_t - \hat{\mathbf{r}}_t^{(S)}. \quad (37)$$

It is not yet measurement noise.

6 Neural signal, artifact, and noise qualification

Noisy ICA identifies structured components but does not know which components are neurons. Qualify each component using:

- spatial localization, compactness, and connectedness;
- annular or membrane-like morphology;
- local temporal coherence;
- event-versus-quiet robust modulation;
- recurrence across windows;
- alignment and stability across seeds and patch offsets;
- correlation with spatial gradients or motion fields;
- saturation/bright-artifact overlap;
- posterior denoising gain and residual reduction.

The output class is one of

$$\{\text{accepted_neural}, \text{structured_artifact}, \text{unresolved}\}. \quad (38)$$

Only accepted components form $\hat{S}$. Coherent rejected components form $\hat{A}$.

The remaining residual may be called $\hat{N}$ only if it exhibits:

- low nonzero-lag temporal autocorrelation;
- low spatial correlation beyond the optical scale;
- little event-triggered energy at known neurons;
- low correlation with motion/edge maps;
- stable quiet statistics;
- variance consistent with intensity.

Fluorescence measurements often contain signal-dependent shot noise plus approximately signal-independent detector noise; intensity-conditioned variance or a variance-stabilized lane should therefore be considered rather than an unqualified homoscedastic Gaussian assumption.

7 Why independence alone is insufficient

A naive signal-noise independence objective permits trivial decompositions such as $S = R, N = 0$ or $S = 0, N = R$. The objective must therefore combine independence with explicit structural and residual constraints. A conceptual criterion is

$$\begin{aligned} \mathcal{J} = & D_{\text{ind}}(s_1, \dots, s_K) + \lambda_{SN} D_{\text{ind}}(S, N) \\ & + \lambda_Q D(N, N_{\text{quiet}}) + \lambda_W R_{\text{white}}(N) + \lambda_A R_{\text{spatial}}(A, S) + \lambda_C E_{\text{closure}}. \end{aligned} \quad (39)$$

The first implementation should not attempt one opaque monolithic loss. It should expose the effects of noise-corrected whitening, noisy density modeling, posterior denoising, and component qualification as separate ablations.

8 Stability requirements

8.1 Stage-1 numerical stability

Require:

- covariance/eigenvalue floors and condition-number limits;
- bounded demixing angle changes;
- sign/permutation continuity;
- gradient clipping and rollback;
- finite dictionaries and bandwidth bounds;
- unresolved fallback rather than forced classification.

8.2 Stage-2 numerical stability

Require:

- positive-semidefinite noise-corrected covariance;
- stable rank across temporal blocks;
- projected-noise variance floors and ceilings;
- demixer decorrelation/conditioning checks;
- log-domain Parzen computations;
- dictionary occupancy/effective-sample diagnostics;
- patch overlap-add closure.

8.3 Scientific stability

A numerically converged model is scientifically unstable if small changes in seed, bandwidth, rank, patch offset, or temporal block produce different neurons or move large event energy among B, S, A, N . Report:

- matched component-map correlations;
- demixer subspace angles;
- candidate Jaccard;
- accepted-component Jaccard;
- event-preservation variation;
- leakage-matrix confidence intervals;
- unresolved/fallback rate.

9 Real-time architecture

Full global Parzen ICA refitting at every frame is neither necessary nor advisable. Use two timescales.

9.1 Fast path

For every frame:

1. form the Stage-1 pair;
2. apply the accepted 2×2 demixer;
3. reconstruct/subtract background;
4. extract overlapping Stage-2 patches;
5. apply fixed local projections/demixers;
6. evaluate bounded Parzen responsibilities and posterior means;
7. overlap-add accepted signal;
8. compute residual diagnostics and event features.

9.2 Slow path

At quiet or scheduled intervals:

1. update the Stage-1 demixer;
2. update local Stage-2 demixers;
3. update bounded dictionaries;
4. update the noise model;
5. validate stability;
6. accept or roll back the checkpoint.

The Spon recording period is 20 ms. Real-time candidacy requires the complete frozen inference path to remain below that deadline at the 95th percentile with headroom, while adaptation latency and frequency are reported separately.

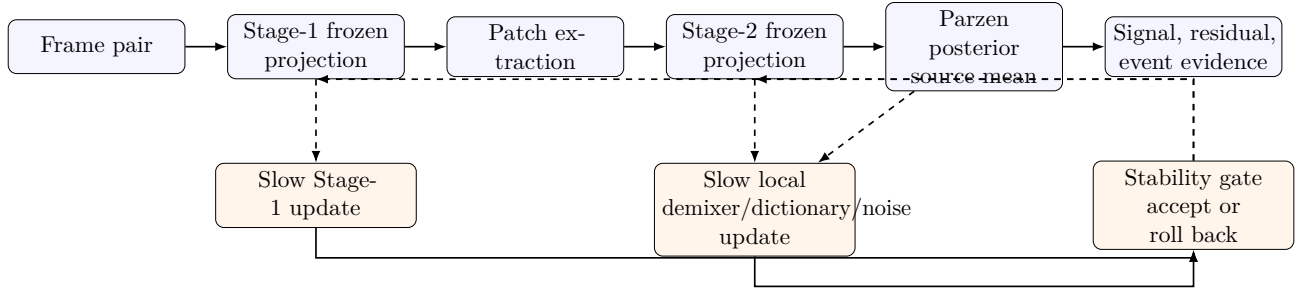


Figure 4: Two-timescale implementation. The fast path performs fixed inference; slow updates are checkpointed, tested, and either accepted or rolled back.

10 Falsification-oriented evaluation

10.1 Synthetic and semi-synthetic construction

Generate exact channels B, S, A, N using:

- static, gain-varying, linearly drifting, and nonlinear backgrounds;
- localized disks and membrane-like annuli;
- fast transients, slow ramps, plateaus, and overlapping events;
- independent and synchronous sources;
- translations, edge motion, saturation, and impulsive artifacts;
- white, correlated, heteroscedastic, and Poisson–Gaussian noise;
- patch-boundary and image-boundary cases.

Semi-synthetic experiments inject exact sources and artifacts into real quiet NeuRev frames while retaining every ground-truth channel.

10.2 Decomposition metrics

The closure error is

$$\mathcal{E}_{\text{closure}} = \frac{\|Y - \hat{B} - \hat{S} - \hat{A} - \hat{N}\|_F^2}{\|Y\|_F^2 + \epsilon}. \quad (40)$$

For true channel C_i and estimated channel $\hat{C}_j$, report a leakage matrix

$$L_{ij} = \frac{\|\text{Proj}_{\hat{C}_j}(C_i)\|_F^2}{\|C_i\|_F^2 + \epsilon}, \quad (41)$$

plus direct NMSE and aligned correlation because projection metrics can favor high-rank channels.

10.3 Event preservation

For each known or injected event, report:

- amplitude ratio and temporal-area ratio;
- onset, peak, and duration error;
- centroid error and footprint/annulus overlap;
- energy moved into background, artifact, and noise;
- local signal-to-quiet evidence before and after each stage.

10.4 Residual validity

Report temporal ACF, spatial correlation versus distance, power spectrum, intensity-conditioned variance, tail behavior, event-triggered residual average, and correlation with gradients/motion. A residual that contains coherent membrane structure is missed signal; a residual with moving paired edges is artifact; a residual with smooth global fluctuations is missed background.

10.5 Downstream NeuRev metrics

Use identical temporal pooling, quiet calibration, NMS, matching, and candidate budgets across lanes. Compare both recall-versus-candidate burden and fixed-budget recall. Unmatched candidates remain unknown. Add annular/footprint-aware matching and pre-NMS versus post-NMS audits so representation failure is separated from detector failure.

11 Mandatory result figures

The implementation package requires the following outputs:

Figure	Required content
Decomposition sheet	Raw, background, Stage-1 residual, neural signal, artifact, noise, sum reconstruction, closure error at quiet/event/shared-miss/artifact frames.
Stage-1 geometry	Pair cloud, common/difference directions, fitted axes, classification confidence, angle and identity tracking.
Staticness scatter	First- vs second-derivative energy, spatial broadness, common-direction cosine, resolved/unresolved labels.
Noise spectrum	Residual covariance, estimated noise, positive $\Sigma_r - \Sigma_n$ spectrum, selected rank and block stability.
Parzen diagnostics	Dictionary centers, clean and noise-convolved densities, projected noise, score, posterior shrinkage curve.
Component galleries	Spatial maps, noisy/clean traces, morphology, motion correlation, stability, acceptance reason.
Leakage matrix	True $B/S/A/N$ versus estimated $B/S/A/N$ with uncertainty.
Residual diagnostics	ACF, spatial correlation, PSD, intensity-variance, event-triggered residual, gradient/motion correlation.
Event preservation	Amplitude/area/timing/morphology/leakage distributions and paired shared-miss traces.
Detection Pareto	Known-label recall versus candidate burden and fixed-budget recall for all anchors and proposed lanes.
Shared failures	Observation-by-method matrix separating representation, threshold, NMS, matching, annotation, and no-evidence failures.
Stability	Demixer angles, component correlations, candidate Jaccard, leakage variation over seeds/windows/patch offsets/ranks.
Latency	Stage and total p50/p95/p99, memory, adaptation frequency, fallback count, and 20 ms deadline.

Table 1: Mandatory visual evidence. Conceptual diagrams in this manuscript are not substitutes for experiment-generated figures.

12 Stage gates

- C1: Implementation integrity.** Deterministic tests, exact closure, collision-safe artifacts, correct frames/coordinates, and Raw Direct anchor.
- C2: Stage-1 identifiability and stability.** Finite conditioning, continuous component identity, bounded updates, and unresolved fallback.
- C3: Stage-1 scientific validity.** Low signal leakage into background, background reduction beyond fixed subtraction, and slow-event preservation.
- C4: Stage-2 numerical stability.** PSD covariance, stable rank, bounded noise variance/dictionaries, stable demixing, overlap-add closure.
- C5: Stage-2 decomposition validity.** Better source recovery than noise-agnostic ICA and a less structured residual without artifact relabeling.
- C6: Real signal preservation.** Stable amplitude, area, timing, morphology, and localization on real known events and reviewed failures.
- C7: Downstream value.** Improved held-out recall in at least three of four bursts or at least 20% candidate reduction without recall loss.

C8: Real-time candidacy. Frozen causal inference below 20 ms at p95, with safe slow-update freeze/rollback behavior.

No later gate is valid when an upstream gate is incomplete.

13 Principal risks

Risk	Failure mechanism	Required response
Slow signal absorbed as background	Low derivative energy resembles persistent background	Staticness margin, spatial/morphology terms, unresolved state, semi-synthetic slow ramps, alternating refinement.
Stage-1 error propagation	Removed signal is unavailable to Stage 2	Preserve raw/background/residual, measure leakage, confidence-weighted/fallback lane, optional one-pass refinement.
Noise treated as source	Gaussian noise is rotationally ambiguous	Explicit noise covariance, noise-corrected subspace, posterior source estimator, residual qualification.
Structured artifact accepted	Motion/saturation are non-Gaussian and coherent	Artifact class, motion/gradient/saturation features, review gallery.
Parzen collapse or saturation	Bandwidth/dictionary/update degeneracy	Bounds, leave-one-out, dictionary diagnostics, finite-difference tests, rollback.
High-rank instability	Too many components and weak identifiability	Local patches, bounded rank, seed/window stability, stop gates.
False confidence from detection	More candidates can raise recall	Fixed-budget comparison, sparse-positive semantics, shared-failure audit.
Real-time drift	Continual adaptation changes component meaning	Fast frozen path, slow checkpointed updates, sign/permutation tracking, fallback.

Table 2: Dominant failure modes and the corresponding implementation safeguards.

14 Conclusion

The proposed hierarchy is viable as a bounded research program, not as an assumption that two consecutive ICA calls automatically identify background, neurons, and noise. Stage 1 should use Parzen ICA to estimate a common/background mode and reconstruct an amplitude-preserving residual. Stage 2 should use local multichannel patches, an explicit noise covariance, a source density convolved with the projected noise distribution, a stochastic score-function update, and posterior source denoising. Coherent components must then be classified as neural, artifact, or unresolved before the final residual is called measurement noise.

The information-theoretic contribution is precise: Parzen densities provide a nonparametric score for adaptive separation; stochastic information-gradient updates provide a bounded online route; and noise convolution converts the same source model into a denoising posterior. The scientific contribution depends on closure, leakage, event preservation, residual validity, stability, and real-time behavior—not on entropy reduction alone.

References

- [1] A. J. Bell and T. J. Sejnowski, “An information-maximization approach to blind separation and blind deconvolution,” *Neural Computation*, vol. 7, no. 6, pp. 1129–1159, 1995.
- [2] D. Erdogmus, K. E. Hild II, and J. C. Principe, “Online entropy manipulation: Stochastic information gradient,” *IEEE Signal Processing Letters*, vol. 10, no. 8, pp. 242–245, 2003.
- [3] D. Erdogmus, K. E. Hild II, J. C. Principe, M. Lázaro, and I. Santamaría, “Adaptive blind deconvolution of linear channels using Rényi’s entropy with Parzen window estimation,” *IEEE Transactions on Signal Processing*, vol. 52, no. 6, pp. 1489–1498, 2004.
- [4] K. E. Hild II, D. Erdogmus, and J. C. Principe, “An analysis of entropy estimators for blind source separation,” *Signal Processing*, vol. 86, no. 1, pp. 182–194, 2006.
- [5] J.-M. Yang and H. Sakai, “A new adaptive filter algorithm for system identification using independent component analysis,” *IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences*, vol. E90-A, no. 8, pp. 1549–1554, 2007.
- [6] J. R. Voss, M. Belkin, and L. Rademacher, “A pseudo-Euclidean iteration for optimal recovery in noisy ICA,” in *Advances in Neural Information Processing Systems 28*, pp. 2872–2880, 2015.
- [7] E. A. Pnevmatikakis *et al.*, “Simultaneous denoising, deconvolution, and demixing of calcium imaging data,” *Neuron*, vol. 89, no. 2, pp. 285–299, 2016.
- [8] J. Friedrich, P. Zhou, and L. Paninski, “Fast online deconvolution of calcium imaging data,” *PLoS Computational Biology*, vol. 13, no. 3, e1005423, 2017.
- [9] E. K. Buchanan *et al.*, “Penalized matrix decomposition for denoising, compression, and improved demixing of functional imaging data,” *arXiv preprint arXiv:1807.06203*, 2018.